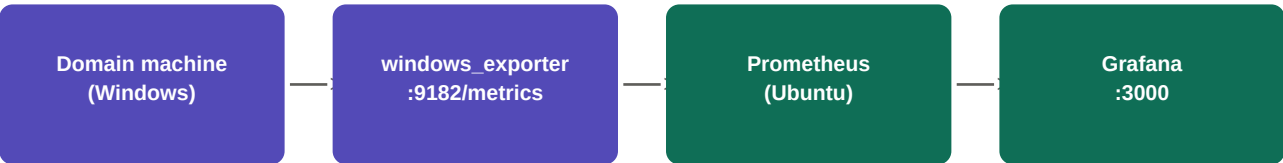


Monitoring a Windows Domain Machine

with Prometheus & Grafana on Ubuntu

Step-by-step setup guide | Ubuntu 22.04 / 24.04 | windows_exporter + Prometheus + Grafana



Overview

This guide walks through setting up a complete monitoring stack for a Windows domain machine. The Windows machine runs a small agent called windows_exporter that exposes system metrics. A separate Ubuntu server runs Prometheus (which collects those metrics) and Grafana (which displays them as dashboards).

Component	Role
windows_exporter	Runs on the Windows machine. Exposes CPU, memory, disk, network and service metrics on port 9182.
Prometheus	Runs on Ubuntu. Scrapes metrics from windows_exporter every 15 seconds and stores them.
Grafana	Runs on Ubuntu. Connects to Prometheus and displays dashboards in the browser on port 3000.

Note: Ubuntu 26.04 does not yet exist. The steps below are tested on Ubuntu 22.04 LTS and 24.04 LTS.

1

Install windows_exporter on the Windows domain machine

All commands in this step are run on the Windows machine you want to monitor, using PowerShell as Administrator.

1.1 Download the installer

Go to the windows_exporter releases page and download the latest .msi file for amd64:

https://github.com/prometheus-community/windows_exporter/releases

1.2 Install silently with the required collectors

```
msiexec /i windows_exporter-0.27.2-amd64.msi ^
ENABLED_COLLECTORS="cpu,cs,logical_disk,net,os,process,service,memory" ^
LISTEN_PORT=9182 /qn
```

This installs windows_exporter as a Windows service that starts automatically on boot.

1.3 Verify it is running

```
Invoke-WebRequest http://localhost:9182/metrics | Select-Object -First 5
```

You should see lines starting with # HELP and # TYPE followed by metric values.

1.4 Open the firewall

Allow the Ubuntu server to reach port 9182 through the Windows firewall:

```
New-NetFirewallRule -DisplayName "Prometheus windows_exporter" ^
-Direction Inbound -Protocol TCP -LocalPort 9182 -Action Allow
```

From the Ubuntu server, confirm connectivity with:

```
curl http://:9182/metrics
```

Note: Replace with the actual IP address of your Windows machine throughout this guide.

2 Install Prometheus on the Ubuntu server

All commands from this step onward are run on the Ubuntu monitoring server.

2.1 Create a dedicated system user

```
sudo useradd --no-create-home --shell /bin/false prometheus
```

2.2 Download and extract Prometheus

Check <https://prometheus.io/download> for the latest version number, then run:

```
cd /tmp
wget https://github.com/prometheus/prometheus/releases/download/v2.52.0/prometheus-2.52.0.linux-amd64.tar.gz
tar xvf prometheus-2.52.0.linux-amd64.tar.gz
cd prometheus-2.52.0.linux-amd64
```

2.3 Install binaries and directories

```
sudo cp prometheus /usr/local/bin/
sudo cp promtool /usr/local/bin/
sudo mkdir /etc/prometheus /var/lib/prometheus
sudo cp -r consoles/ console_libraries/ /etc/prometheus/
sudo chown -R prometheus:prometheus /etc/prometheus /var/lib/prometheus
```

3 Configure Prometheus to scrape the Windows machine

Create the Prometheus configuration file at /etc/prometheus/prometheus.yml:

```
sudo nano /etc/prometheus/prometheus.yml
```

Paste the following content (replace with your actual IP address):

```
global:
  scrape_interval: 15s
  evaluation_interval: 15s

scrape_configs:
  - job_name: 'windows_domain_machine'
    static_configs:
      - targets: ['192.168.1.100']
        labels:
          hostname: 'DOMAIN-PC-01'
```

Validate the configuration file before proceeding:

```
promtool check config /etc/prometheus/prometheus.yml
```

If the output says "SUCCESS", the file is valid.

Note: The hostname label is optional but helps identify the machine in Grafana when you monitor multiple targets.

4 Run Prometheus as a systemd service

Create the systemd service file:

```
sudo nano /etc/systemd/system/prometheus.service
```

Paste the following:

```
[Unit]
Description=Prometheus
After=network.target

[Service]
User=prometheus
ExecStart=/usr/local/bin/prometheus \
  --config.file=/etc/prometheus/prometheus.yml \
  --storage.tsdb.path=/var/lib/prometheus \
  --web.listen-address=0.0.0.0:9090 \
  --storage.tsdb.retention.time=30d
Restart=on-failure

[Install]
WantedBy=multi-user.target
```

Enable and start the service:

```
sudo systemctl daemon-reload
sudo systemctl enable --now prometheus
sudo systemctl status prometheus
```

Open <http://:9090/targets> in your browser. The `windows_domain_machine` target should show a green UP status. If it shows DOWN, check that port 9182 is open on the Windows firewall and that the IP address in `prometheus.yml` is correct.

5 Install Grafana on the Ubuntu server

```
sudo apt install -y apt-transport-https software-properties-common

wget -q -O - https://packages.grafana.com/gpg.key | sudo apt-key add -
echo "deb https://packages.grafana.com/oss/deb stable main" | \
  sudo tee /etc/apt/sources.list.d/grafana.list

sudo apt update
sudo apt install -y grafana
sudo systemctl enable --now grafana-server
```

Grafana is now accessible at <http://:3000>. The default login credentials are username `admin` and password `admin`. You will be prompted to set a new password on first login.

6 Connect Grafana to Prometheus

Follow these steps inside the Grafana web interface:

- Log in at <http://:3000>
- Go to Connections then Data sources then Add data source
- Select Prometheus from the list
- Set the URL field to <http://localhost:9090>
- Scroll down and click Save and Test

A green banner saying "Data source is working" confirms the connection is successful.

7 Import the Windows monitoring dashboard

Instead of building a dashboard from scratch, import the community dashboard ID 14694, which is purpose-built for `windows_exporter` and covers all major metrics out of the box.

- Go to Dashboards then Import
- Type 14694 in the ID field and click Load
- Select your Prometheus data source from the dropdown
- Click Import

The dashboard includes the following panels:

Panel	What it shows
CPU usage %	Overall processor load across all cores
Memory used / free	Physical RAM consumption and availability

Disk read / write	I/O throughput per logical disk volume
Network in / out	Bytes per second on each network adapter
Running services	State of Windows services (running or stopped)

8

Useful PromQL queries for custom panels

Use these queries when building additional panels in Grafana. Enter them in the panel editor under the Query tab with Code mode selected.

CPU usage percentage

```
100 - (avg by(instance)(rate(windows_cpu_time_total{mode="idle"}[5m])) * 100)
```

Memory used percentage

```
100 - (windows_os_physical_memory_free_bytes / windows_cs_physical_memory_bytes * 100)
```

Disk free percentage on drive C

```
windows_logical_disk_free_bytes{volume="C:"} / windows_logical_disk_size_bytes{volume="C:"} * 100
```

Network received (bytes per second)

```
rate(windows_net_bytes_received_total[5m])
```

Check if a specific service is running

```
windows_service_state{name="wuauserv", state="running"}
```

A value of 1 means the service is running. A value of 0 means it is stopped. Replace wuauserv with any Windows service name.

Troubleshooting

Symptom	Likely cause and fix
Target shows DOWN in Prometheus	Port 9182 is blocked. Check the Windows firewall rule and confirm the IP address in prometheus.yml is correct.
No data in Grafana panels	The data source URL may be wrong. Confirm it is set to http://localhost:9090 and click Save and Test again.
windows_exporter service not starting	Check the Windows Event Viewer under Application logs for errors. Re-run the msixec command as Administrator.
Grafana shows N/A for all metrics	The scrape interval may not have elapsed yet. Wait 30 seconds and refresh. Also check the Prometheus Targets page.

End of guide